

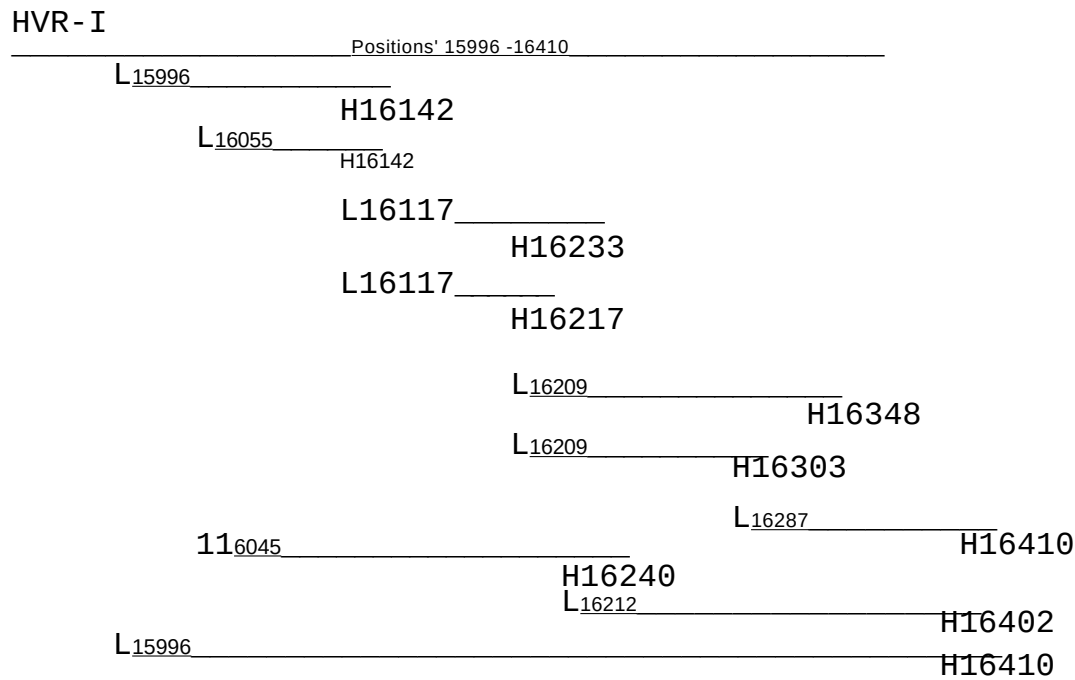


Figure 2: System of primers for HVR-I amplification and sequencing

Typing of 22 coding region SNPs (GenoCore22 reaction)

The assessment of mtDNA haplogroups based on HVR-I sequences was refined and/or confirmed by typing 22 haplogroup diagnostic SNPs in the coding region of the mtDNA genome.

SNaPshot minisequencing has been previously applied to type SNPs in the mitochondrial genome (Brandstatter et al., 2003; Brandstatter et al., 2006; Grignani et al., 2006; Niederstatter et al., 2006; Haak et al., 2010), on autosomal chromosomes (Grimes et al., 2001; Dixon et al., 2005; Sanchez et al., 2006; Bouakaze et al., 2007) and on the Y-chromosome (Sanchez et al., 2003; Sanchez et al., 2005; Brion et al., 2005a; Brion et al., 2005b; Onofri et al., 2006; Bouakaze et al., 2009). These studies include work on modern, forensic and ancient DNA and have demonstrated the robustness, reliability and sensitivity of the SNaPshot minisequencing approach (Bouakaze et al., 2009).

In general, coding region SNPs were characterised using a multiplex PCR followed by a multiplexed Single Base Extension (SBE) reaction (GenoCore22, Haak et al., 2010) using the SNaPshot minisequencing technique (Applied Biosystems). This approach consists of a preliminary multiplex PCR amplification, in the present